Active Material Screening via Multi-Armed Bandits

Accelerating chemical synthesis by replacing exhaustive trial-and-error with uncertainty-aware reinforcement learning.

WET-LAB BOTTLENECK

Synthesis and testing are cost-intensive and slow. Exhaustive search fails.

DECISION GOAL

Maximize discovery yield under a strict budget of 1,000 experimental runs.



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METHODOLOGY & FORMULATION

Stochastic Multi-Armed Bandit & UCB1

Environment

20 candidate alloys with true performance scores μ in [0.10, 0.95] corrupted by 5% Gaussian noise.

Action & Reward

Select candidate arm a_t ∈ {1..20}. Reward returns noisy measured laboratory yield r_t = μ + ε.

UCB1 Decision Rule

$$a_t = \arg\max\left[\hat{Q}_i + c\sqrt{\frac{\ln t}{N_i}}\right]$$

Optimism under uncertainty balancing exploration bonus with empirical mean.

Random Baseline

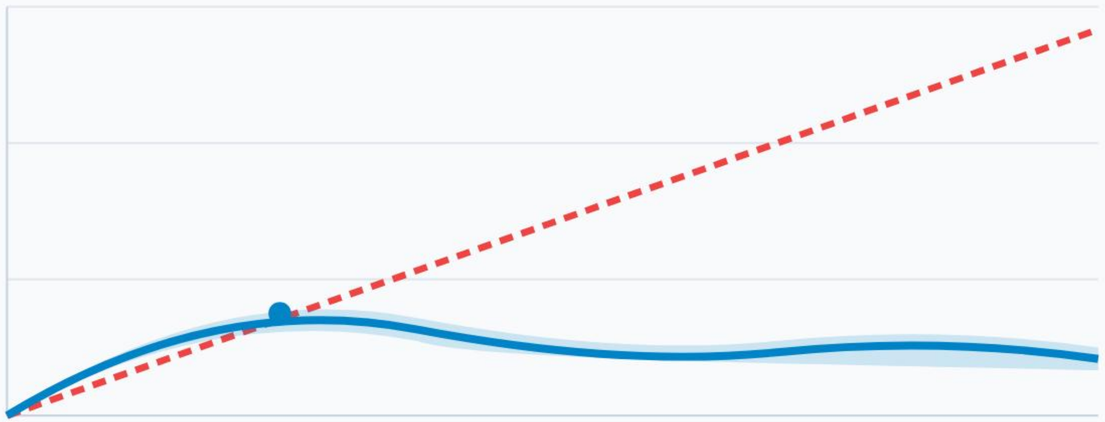
Uniform random allocation over the 20 candidates, modeling unguided screening runs.

Experimental Protocol: 100 Independent Trials · Horizon T = 1,000 Steps · Metric: Regret & Yield

Exploration constant c = √2

Learning Curves: UCB1 vs. Random Screening

Cumulative Regret Trajectory (T = 1,000 Steps)



t = 0 Screening Rounds t = 1,000

SAMPLE EFFICIENCY

< 15% Budget

UCB1 isolates optimal candidate (μ* = 0.95) within ~150 experimental pulls.

ASYMPTOTIC BEHAVIOR

O(ln T) vs. O(T)

Logarithmic regret bound eliminates persistent trial loss observed in random search.

NOISE RESILIENCE

σ = 0.05 Tolerance

Confidence bounds prevent deception from stochastic measurement error spikes.

● UCB1 Agent (Mean of 100 trials) — Random Selection Baseline

SYSTEM BOUNDARIES

Real-World Limitations

Discrete vs Continuous Space

Real alloys exist in continuous phase diagrams with infinite compositions.

Non-Stationary Noise

Laboratory sensors suffer calibration drift and heteroscedastic batch errors.

Asymmetric Cost Overhead

Precursor scarcity and reaction runtimes make experimentation costs non-uniform.

Future Work: Extend formulation to Contextual Bandits and Gaussian Process Bayesian Optimization to handle continuous formulation vectors.

Ethics & Broader Impact

Dual-Use & Toxic Hazards

Unconstrained yield maximization risks synthesizing hazardous or toxic byproducts without penalty checks.

Diversity vs Exploitation Bias

Aggressive exploitation around early successes risks overlooking structurally novel material classes.

Human-in-the-Loop Validation

Automated robotic synthesis loops must be gated by safety layers and expert verification protocols.

Image Sources



https://cdn.labmanager.com/assets/image/55313/high-throughput-screening-and-robotics-s.webp

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